Doubly Robust Estimation of Counterfactual Policy Mean Embeddings

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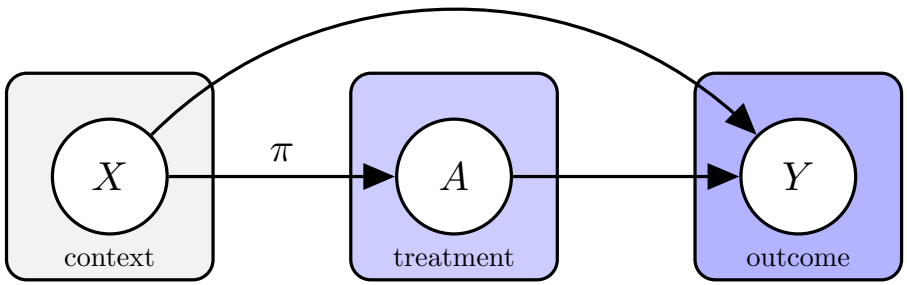
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Setting

We are given an observational dataset $\mathcal{D}_0 = (x_i, a_i, y_i)_{i=1,\dots,n}$, for each interaction the system:

- observed a context $x_i \in \mathcal{X}$, and $x_i \sim P_X$,
- sampled an action (treatment) $a_i \in \mathcal{A}$ from a logging policy π_0 with propensity $\pi_{i,0} = \pi_0(a_i|x_i)$,
- observes a loss (outcome) $y_i \in \mathcal{Y}$ and $y_i \sim P_Y(\cdot|x_i, a_i)$

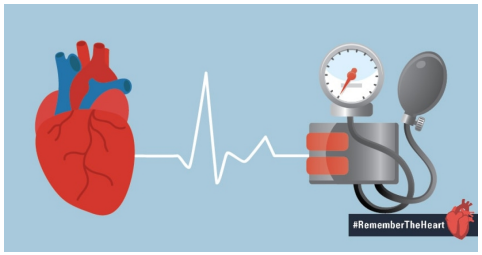


Counterfactual question: *"What would have been the outcome of the experiment, should we treat with stochastic intervention π'?"*

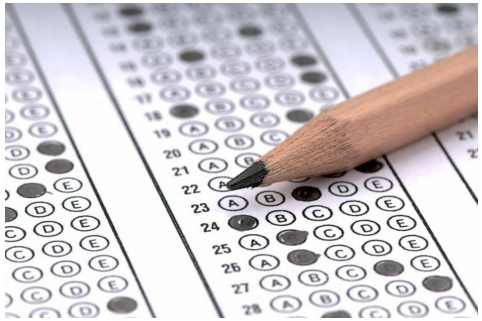
Outcomes

Traditionally, causal inference and policy evaluation target **scalar outcomes**:

- Health: blood pressure, cholesterol level, survival time.
- Economics: income, employment status, GDP growth.
- Education: test scores, years of schooling.







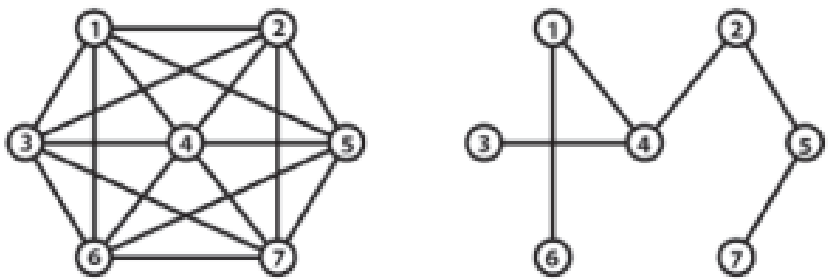
Structured Outcomes: Medical Imaging

- Treatment/Policy: choice of treatment (choice of drug, dosage of the drug).
- Outcome: **follow-up MRI/CT scan**.
- Goal: compare counterfactual distributions of lesion shapes, sizes, textures.



Structured Outcomes: Graphs

- Treatment/Policy: recommendation algorithm (who to follow, what to read).
- Outcome: **user's social network graph**.
- Question: do policies shift network topology (e.g., dense, sparse)?



Dense Sparse

Structured Outcomes: Text

- Treatment/Policy: which message / text the system shows to the user.
- Outcome: **the user's written reply**.
- Goal: evaluate how alternative policies shift the distribution of replies (tone, length, engagement, sentiment).

Policy π (Neutral messages)
Prompt: *"Would you like to hear more about healthy diets?"*
User reply: *"Maybe, I'm not sure."*

Policy π' (Engaging messages)
Prompt: *"Discover how small changes in your diet can boost your energy!"*
User reply: *"Sounds interesting, tell me more!"*

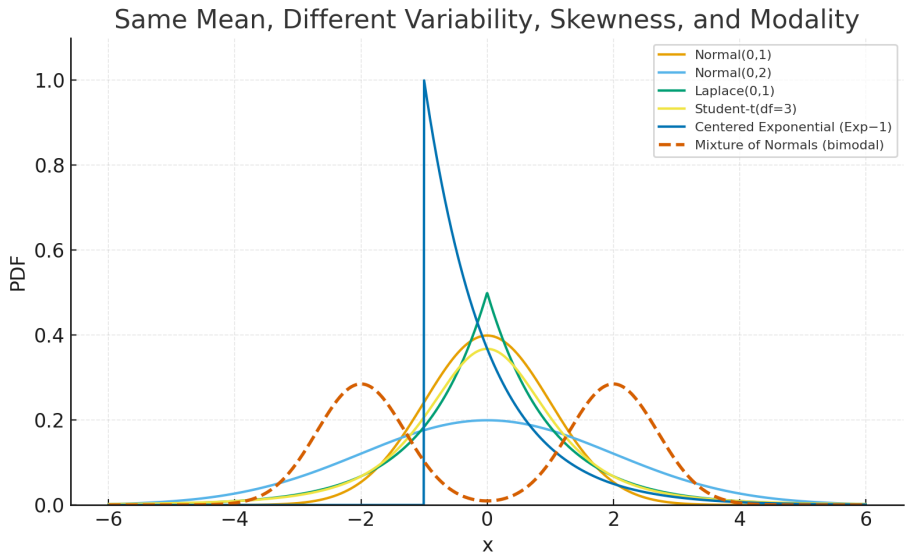
Beyond counterfactual outcome expectations

Classical causal inference tackles the **expected value function**: $\mathbb{E}[Y(\pi)]$: for **scalar outcomes**, this **expectation** often suffices.

However, for **structured outcomes** (e.g. images, text, graphs), averages are meaningless.

Furthermore, for all type of outcomes, the **entire outcome distribution** matters:

- A policy may alter the **shape or tails** of the distribution (e.g. rare adverse events) without necessarily affecting the mean outcome.



We need tools to compare the **entire counterfactual distributions**.

Counterfactual Policy Mean Embedding (CPME)

Let $Y(a)$ the potential outcome under treatment a. A kernel $k_{\mathcal{Y}} : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$ defines an RKHS $\mathcal{H}_{\mathcal{Y}}$.

CPMEs[1] embed the distribution of $\{Y(a)\}_{a\sim\pi}$:

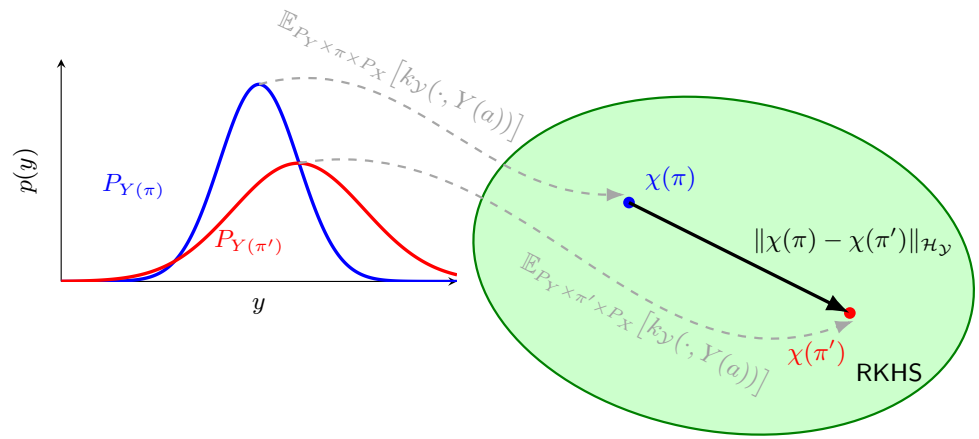
$$\chi(\pi) := \mathbb{E}_{P_Y \times \pi \times P_X}[k_{\mathcal{Y}}(\cdot, Y(a))] \in \mathcal{H}_{\mathcal{Y}}.$$

Key ideas:

- If $k_{\mathcal{Y}}$ is *characteristic*, $\chi(\pi)$ uniquely identifies the interventional π.
- Each π is mapped to a point in $\mathcal{H}_{\mathcal{Y}}$; distances reflect distributional shifts:

$$\|\chi(\pi) - \chi(\pi')\|_{\mathcal{H}_{\mathcal{Y}}} = 0 \Leftrightarrow P_{Y(\pi)} = P_{Y(\pi')}.$$

- Choosing $k_{\mathcal{Y}}$ to encode the geometry of images, text, or graphs (e.g., RBF on embeddings, string/graph kernels) makes CPME suited to structured $\mathcal{Y}$.



[1] Our setting extends Counterfactual Mean Embeddings of Muandet et al. (2021) to off-policy evaluation.

A Plug-in CPME estimator

Assumption (Regularity)

Assume bounded, continuous kernels $k_{\mathcal{A}\mathcal{X}}$ and $k_{\mathcal{Y}}$; measurability; and $k_{\mathcal{Y}}$ characteristic.

- **Conditional mean operator (CMO)**. Define the conditional mean embedding:

$$\mu_{Y|A,X}(a,x) \;:=\; \mathbb{E}\big[k_{\mathcal{Y}}(\cdot,Y) \mid A=a,\; X=x\big] \;\in\; \mathcal{H}_{\mathcal{Y}},$$

  The CME defines a bounded linear operator

$$C_{Y|A,X} : \mathcal{H}_{\mathcal{A}\mathcal{X}} \;\to\; \mathcal{H}_{\mathcal{Y}}, \qquad C_{Y|A,X}\,\phi_{\mathcal{A}\mathcal{X}}(a,x) \;=\; \mu_{Y|A,X}(a,x),$$

  where $\phi_{\mathcal{A}\mathcal{X}}(a,x) = k_{\mathcal{A}\mathcal{X}}(\cdot,(a,x))$ is the joint feature map.

- Then, when introducing the following kernel mean embedding:

$$\mu_{\pi} \;:=\; \mathbb{E}_X\; \mathbb{E}_{a\sim\pi(\cdot|X)}\big[\phi_{\mathcal{A}\mathcal{X}}(a,X)\big] \;\in\; \mathcal{H}_{\mathcal{A}\mathcal{X}},$$

We obtain the following **decoupled formulation of CPME:**

$$\boxed{\chi(\pi) \;=\; C_{Y|A,X}\,\mu_{\pi}} \in\; \mathcal{H}_{\mathcal{Y}}$$

Estimate $\hat{C}_{Y|A,X}$ via operator ridge regression, and $\hat{\mu}_{\pi}$ with a kernel mean embedding estimator, then plug in:

$$\boxed{\hat{\chi}_{\mathrm{pi}}(\pi) \;=\; \hat{C}_{Y|A,X}\,\hat{\mu}_{\pi}} \;\in\; \mathcal{H}_{\mathcal{Y}}.$$

Plug-in Consistency and Rates (SRC & EVD)

Proposition (Plug-in convergence rate in RKHS)

Under selection-on-observables, RKHS regularity, and standard source (smoothness $c \in (1, 3]$) and eigenvalue decay (complexity $b \in (0, 1]$) assumptions, choose

$$\lambda \;=\; n^{-1/(c+1/b)}.$$

Then, with high probability,

$$\left\|\hat{\chi}_{\mathrm{pi}}(\pi) - \chi(\pi)\right\|_{\mathcal{H}_{\mathcal{Y}}} = \mathcal{O}_p\Big(r_C(n)\Big) = \mathcal{O}_p\Big(n^{-\frac{c-1}{2(c+1/b)}}\Big)\ .$$

Interpretation.

- Error is dominated by estimating $C_{Y|A,X}$ (operator regression); $\hat{\mu}_\pi$ attains $n^{-1/2}$.
- b smaller $\Rightarrow$ slower spectral decay (harder problem); c larger $\Rightarrow$ smoother operator (easier).
- Best rate $= n^{-1/4}$ when $c = 3$ (favorable smoothness).

Teaser. A **doubly robust** estimator of CPME achieves root-n inference when nuisances are learned well (next section).

The Efficient Influence Function (EIF) for the CPME

Goal. Construct an estimator of $\chi(\pi)$ that is *first-order unbiased* and *efficient*. Let $\widehat{P}$ denote the empirical law induced by the estimated nuisances $(\hat{C}_{Y|A,X},\, \hat{\mu}_\pi)$. The plug-in estimator $\widehat{\chi}_{\mathrm{pi}}(\pi) = \chi(\pi; \widehat{P})$ then inherits bias from these nuisance estimates.

Intuition: The EIF ψ^π acts as the *gradient* of the functional $\chi(\pi; P)$ with respect to the data-generating law P. Since $\chi(\pi; P)$ is pathwise differentiable:

$$\chi(\pi; P) - \chi(\pi; P_0) = \mathbb{E}_{P_0}\left[\psi^\pi(Z)\, \frac{dP - dP_0}{dP_0}(Z)\right] + o(\|P - P_0\|).$$

Replacing P_0 by $\widehat{P}$ gives

$$\chi(\pi; P_0) - \chi(\pi; \widehat{P}) \;\approx\; \mathbb{E}_{\widehat{P}}[\psi^\pi(Z)],$$

meaning that the EIF ψ^π captures how small estimation errors in the distribution $(\widehat{P} - P_0)$ translate into first-order changes in the parameter.
This expansion motivates the *one-step* or *doubly robust* estimator:

$$\widehat{\chi}_{\mathrm{dr}}(\pi) = \widehat{\chi}_{\mathrm{pi}}(\pi) + \tfrac{1}{n} \sum_{i=1}^{n} \widehat{\psi}^\pi(Z_i).$$

The EIF corrects the first order plug-in bias, yielding a **semiparametric efficient** estimator, achieving the minimum possible asymptotic variance.

From Plug-in to One-Step: Double Robustness and Better Rates

EIF for the CPME.

$$\psi^{\pi}(y, a, x) = \frac{\pi(a \mid x)}{\pi_0(a \mid x)} \{k_{\mathcal{Y}}(\cdot, y) - \mu_{Y|A,X}(a, x)\} + \int \mu_{Y|A,X}(a', x)\pi(da' \mid x) - \chi(\pi).$$

One-step DR estimator (using estimated nuisances):

$$\hat{\chi}_{\mathrm{dr}}(\pi) = \hat{\chi}_{\mathrm{pi}}(\pi) + \frac{1}{n}\sum_{i=1}^{n}\left[\frac{\pi(a_i \mid x_i)}{\hat{\pi}_0(a_i \mid x_i)}\left\{k_{\mathcal{Y}}(\cdot, y_i) - \hat{\mu}_{Y|A,X}(a_i, x_i)\right\} + \int \hat{\mu}_{Y|A,X}(a, x_i)\pi(da' \mid x_i) - \hat{\chi}_{\mathrm{pi}}(\pi)\right].$$

Proposition (DR convergence rate in RKHS)

- **Double robustness.** *Consistent if either $\hat{\pi}_0$ or $\hat{\mu}_{Y|A,X}$ is correct.*
- **Rate improvement.** *In $\mathcal{H}_{\mathcal{Y}}$,*

$$\left\|\hat{\chi}_{\mathrm{dr}}(\pi) - \chi(\pi)\right\| = \mathcal{O}_p\Big(n^{-1/2} + r_{\pi_0}(n)\, r_C(n)\Big).$$

  If $r_{\pi_0}(n) \asymp r_C(n) \asymp n^{-1/4}$, the product is $n^{-1/2} \Rightarrow \sqrt{n}$-rate—improving over the plug-in's $n^{-1/4}$ ceiling.
- **Bias cancellation.** *The IPW residual term removes first-order plug-in bias; the augmentation stabilizes variance and reduces model misspecification error.*
- **Efficiency.** *The EIF achieves the semiparametric efficiency bound (minimal asymptotic variance among regular estimators) under standard conditions.*

Application I: DR-KPT — Testing Counterfactual Distributions

Goal. Compare counterfactual outcome distributions under two policies π, π':
$H_0 : \mathbb{E}_{P_\pi}[k_{\mathcal{Y}}(\cdot, Y)] = \mathbb{E}_{P_{\pi'}}[k_{\mathcal{Y}}(\cdot, Y)]$.

EIF-based moment. Defining $\beta_\pi(x) = \mathbb{E}_{a\sim\pi(\cdot|x)}[\mu_{Y|A,X}(a,x)]$.

$$\varphi_{\pi,\pi'}(y,a,x) = \left(\frac{\pi(a|x)-\pi'(a|x)}{\pi_0(a|x)}\right)\{k_{\mathcal{Y}}(\cdot,y) - \mu_{Y|A,X}(a,x)\} + \beta_\pi(x) - \beta_{\pi'}(x)$$

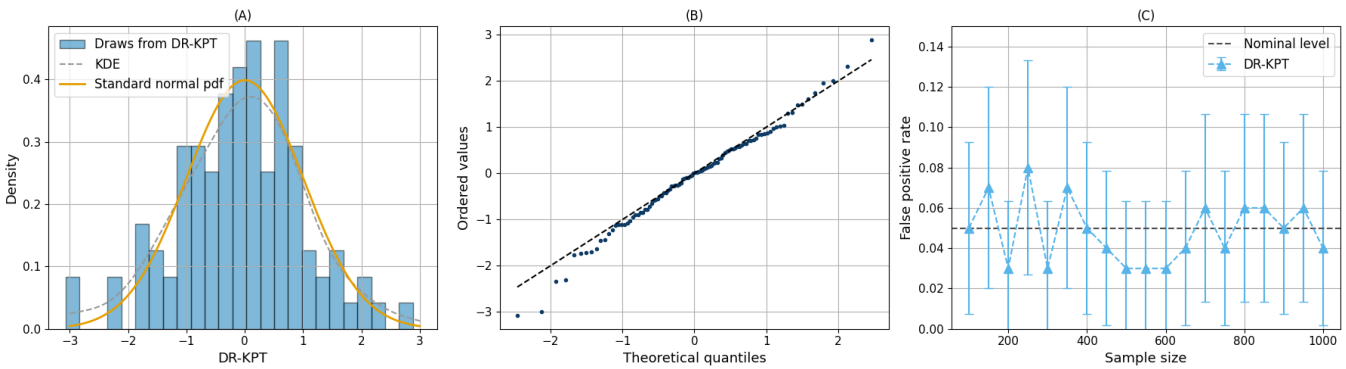
Then $H_0 \iff \mathbb{E}[\varphi_{\pi,\pi'}] = 0$.

Cross U-statistic. Split data in halves; estimate nuisances on one and evaluate on the other $(\hat{\varphi}_i, \tilde{\varphi}_j)$. $f^\dagger_{\pi,\pi'}(Y_i, A_i, X_i) = \frac{1}{n-m}\sum_{j=m+1}^{n}\langle\hat{\varphi}_i, \tilde{\varphi}_j\rangle_{\mathcal{H}_{\mathcal{Y}}}$, $T^\dagger_{\pi,\pi'} = \frac{\sqrt{m}\bar{f}^\dagger}{S^\dagger}$.

Proposition (Asymptotic normality and decision rule)

Under regularity and $r_{\pi_0}(n)r_C(n) = \mathcal{O}(n^{-1/2})$, $T^\dagger_{\pi,\pi'} \Rightarrow \mathcal{N}(0,1)$, $p = 1 - \Phi(T^\dagger_{\pi,\pi'})$; *reject* H_0 *at level* α *if* $p \le \alpha$.

Experiment (Null calibration). DR-KPT correctly maintains type-I error ($\alpha = 0.05$); $T^\dagger_{\pi,\pi'}$ is approximately standard normal.



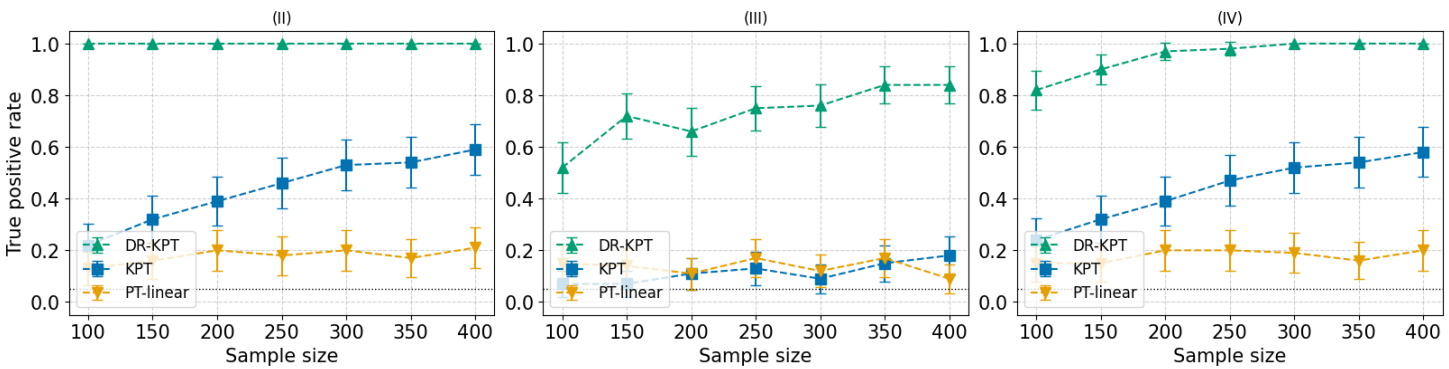
DR-KPT vs. baselines: power under Scenarios II–IV

Baselines.

- **KPT** (adapted from KTE, *estimated* propensities).
- **PT-linear**: linear kernel (tests only mean differences).

Result. DR-KPT shows higher true-positive rates in all alternatives:

- Detects pure mean shifts (Scenario II) and non-mean shifts (bimodality, Scen. III) as well as combined shifts (Scen. IV).
- KPT / PT-linear often lose power due to propensity estimation error or insensitivity to higher moments.
- EIF-based, doubly robust construction $\Rightarrow$ better calibration + power with learned nuisances; cross-U handles MMD degeneracy; computation is efficient.



Application II - Sampling from the counterfactual distribution

Goal. Use the estimated embedding $\hat{\chi}(\pi) \in \mathcal{H}_{\mathcal{Y}}$ to generate pseudo-samples $\tilde{y}_{1:m}$ whose empirical law approximates the outcome distribution under π.

Deterministic kernel herding (from $\hat{\chi}(\pi)$).

$$\tilde{y}_1 := \arg\max_{y \in \mathcal{Y}} \hat{\chi}(\pi)(y), \qquad \tilde{y}_t := \arg\max_{y \in \mathcal{Y}} \left[\hat{\chi}(\pi)(y) - \frac{1}{t}\sum_{\ell=1}^{t-1} k_{\mathcal{Y}}(\tilde{y}_\ell, y)\right], \; t = 2, \ldots, m.$$

Proposition (Guarantees (MMD & weak convergence))

Under mild regularity (locally compact $\mathcal{Y}$, $\mathcal{H}_{\mathcal{Y}} \subset \mathcal{C}_0$):

$$\mathrm{MMD}\left(\tilde{P}^m_{Y,pi}, \nu(\pi)\right) = \mathcal{O}_p\big(r_C(n, b, c) + m^{-1/2}\big),$$

$$\mathrm{MMD}\left(\tilde{P}^m_{Y,dr}, \nu(\pi)\right) = \mathcal{O}_p\big(n^{-1/2} + r_{\pi_0}(n) r_C(n, b, c) + m^{-1/2}\big),$$

$\Rightarrow \tilde{y}_{pi,j} \rightsquigarrow \nu(\pi)$ *and* $\tilde{y}_{dr,j} \rightsquigarrow \nu(\pi)$.

m trades compute for quality; DR embedding improves via the faster $n^{-1/2}$-driven term.

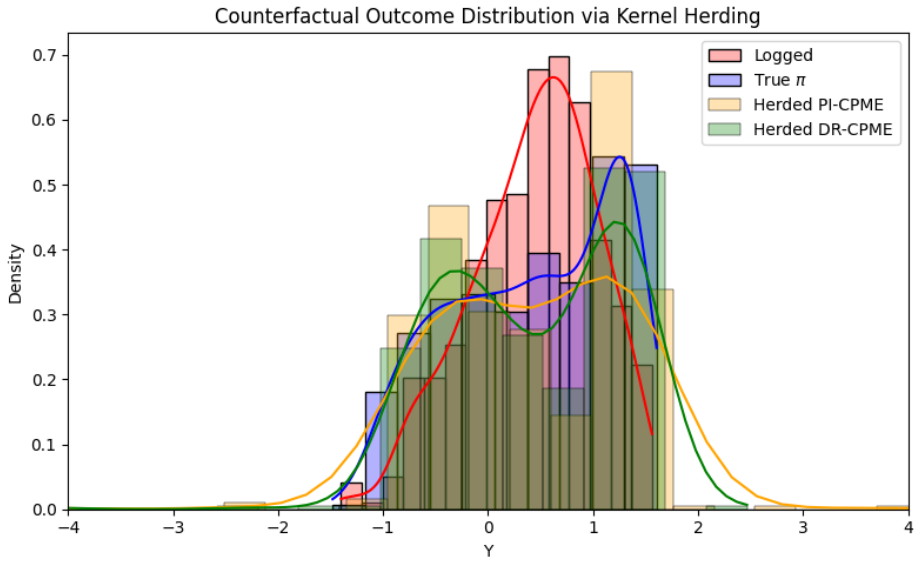
Motivation. In simulation studies, we can arbitrarily sample from any given policy to assess the behavior of our test under controlled conditions.

Application II: Sampling via kernel herding - setup & qualitative result

Goal. Generate counterfactual samples from $\hat{\chi}(\pi) \in \mathcal{H}_{\mathcal{Y}}$ using kernel herding; compare **Plug-in** vs. **DR** CPME.

Scenarios. Logging policy $\pi_0 \in \{\text{uniform, logistic}\}$; outcome $g \in \{\text{quadratic, sinusoidal (nonlinear)}\}$.

- Herd from $\hat{\chi}_{\text{pi}}(\pi)$ and $\hat{\chi}_{\text{dr}}(\pi)$ (Alg. sampling).
- Compare herded empirical law $\tilde{P}_Y^m$ to oracle $\nu(\pi)$ (synthetic ground truth).
- Metrics: Wasserstein distance and MMD (Gaussian kernel).



Logistic π_0, nonlinear outcome: logged vs. oracle vs. Plug-in/DR herded.

Application II: Sampling fidelity - Wasserstein & MMD (herded vs. oracle)

Wasserstein distance

Method	log–nonlin	log–quad	uni–nonlin	uni–quad
Plug-in	1.29e-01 ± 2.6e-01	1.41e-01 ± 4.9e-02	9.08e-02 ± 3.7e-01	6.78e-02 ± 1.9e-02
DR	**8.60e-02** ± 2.2e-02	1.36e-01 ± 3.9e-02	**5.00e-02** ± 1.5e-02	**6.63e-02** ± 1.6e-02

MMD (Gaussian kernel)

Method	log–nonlin	log–quad	uni–nonlin	uni–quad
Plug-in	1.11e-03 ± 5.9e-03	9.85e-04 ± 6.0e-04	1.92e-04 ± 1.2e-03	3.31e-04 ± 2.5e-04
DR	**4.38e-04** ± 3.6e-04	9.80e-04 ± 6.0e-04	**6.49e-05** ± 4.4e-05	3.51e-04 ± 2.5e-04

Takeaways.

- DR herding has lower Wasserstein/MMD, especially under *nonlinear* outcomes and *logistic* logging where misspecification is hardest.
- Corroborates the theory: DR CPME improves the embedding ($n^{-1/2}$-driven term), yielding higher-fidelity samples.

Discussion & Applications

Summary. We introduced the **Counterfactual Policy Mean Embedding (CPME)** with an analysis of plug-in and **EIF-based doubly robust** estimation.

Applications.

- **Testing:** DR-KPT detects distributional shifts between policies with calibrated type-I error and high sensitivity.
- **Sampling:** deterministic herding from $\hat{\chi}(\pi)$ generates pseudo-outcomes approximating the counterfactual distribution, with higher fidelity under DR estimation.

Limitations.

- Standard identification assumptions (consistency, exchangeability, positivity).
- RKHS regularity and kernel scalability ($O(n^2)$).

Future work.

- *Distributional policy optimization* under risk or safety constraints.
- *Robustness extensions*: sensitivity analysis, proximal and IV formulations, partial identification for $\chi(\pi)$.

Ongoing direction: extending kernel treatment effects to **adaptively collected data**.
Preprint: `https://arxiv.org/pdf/2510.10245`

References

 Muandet, Krikamol et al. (2021). "Counterfactual mean embeddings". In: *Journal of Machine Learning Research* 22.162, pp. 1–71.

Appendix: Plug-in Estimator: closed form solutions

Goal. Estimate $C_{Y|A,X}$ and μ_π, then plug in:

$$\boxed{\hat{\chi}_{\mathrm{pi}}(\pi) \;=\; \hat{C}_{Y|A,X}\,\hat{\mu}_\pi} \;\in\; \mathcal{H}_{\mathcal{Y}}.$$

Estimate $\hat{C}_{Y|A,X}$ (operator ridge). For $\lambda > 0$,

$$\hat{\mathcal{L}}^c(C) \;=\; \frac{1}{n}\sum_{i=1}^{n} \left\| k_{\mathcal{Y}}(\cdot, y_i) \;-\; C\, k_{\mathcal{A}\mathcal{X}}(\cdot, (a_i, x_i)) \right\|^2_{\mathcal{H}_{\mathcal{Y}}} + \lambda \| C \|^2_{HS}.$$

Let $\hat{C}_{Y|A,X} = \arg\min_C \hat{\mathcal{L}}^c(C)$. By kernelization,

$$\hat{\mu}_{Y|A,X}(a, x) \;=\; \hat{C}_{Y|A,X}\, k_{\mathcal{A}\mathcal{X}}(\cdot, (a, x)) \;=\; \sum_{i=1}^{n} \beta_i(a, x)\, k_{\mathcal{Y}}(\cdot, y_i),$$

$$\boldsymbol{\beta}(a, x) \;=\; \left(K_{AX,AX} + n\lambda I\right)^{-1} K_{AX,ax},$$

Estimate $\hat{\mu}_\pi$.

$$\text{Discrete } \mathcal{A}: \quad \hat{\mu}_\pi = \frac{1}{n}\sum_{i=1}^{n}\sum_{a\in\mathcal{A}} \pi(a \mid x_i)\, k_{\mathcal{A}\mathcal{X}}(\cdot, (a, x_i)).$$

$$\text{Continuous } \mathcal{A}: \quad \hat{\mu}_\pi = \frac{1}{n}\sum_{i=1}^{n}\frac{1}{M}\sum_{m=1}^{M} k_{\mathcal{A}\mathcal{X}}(\cdot, (a_{im}, x_i)),\; a_{im} \sim \pi(\cdot \mid x_i).$$

Appendix: The Efficient Influence Function (EIF) for the CPME – I

Pathwise view. Consider a regular submodel $\{P_\varepsilon : \varepsilon \in \mathbb{R}\}$ through the true law P_0, with density

$$p_\varepsilon(z) = p_0(z)\{1 + \varepsilon s(z)\} + o(\varepsilon),$$

where $s(Z) \in L^2(P_0)$ is the **score function**. It encodes a valid infinitesimal perturbation of P_0: the directions in which the data law can move smoothly.

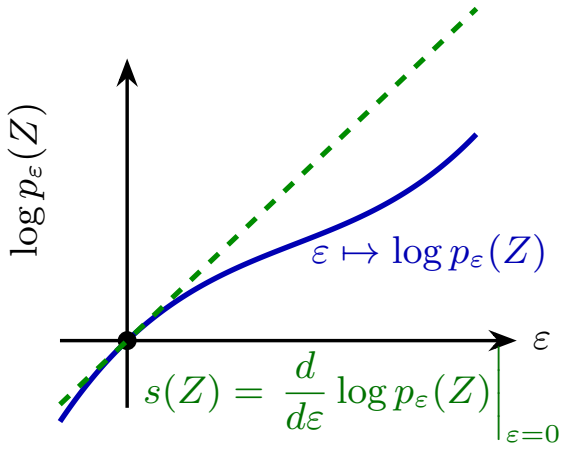
The parameter path $\varepsilon \mapsto \chi_\varepsilon(\pi) \in \mathcal{H}_\mathcal{Y}$ is **pathwise differentiable** if for every score s,

$$D_s\chi(\pi) := \left.\frac{d}{d\varepsilon}\chi_\varepsilon(\pi)\right|_0 \text{ exists in } \mathcal{H}_\mathcal{Y}.$$

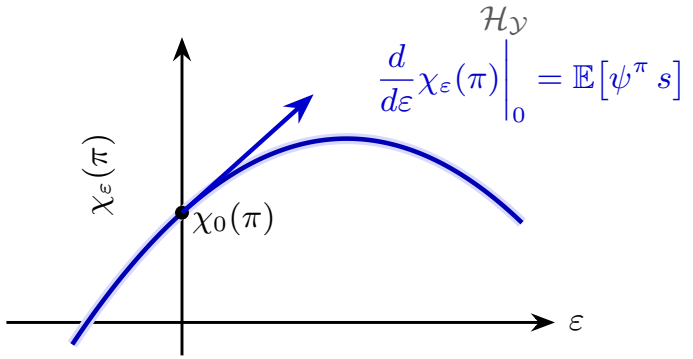
By the Riesz representation theorem, there exists a unique $\psi^\pi(Y, A, X) \in L^2(P_0; \mathcal{H}_\mathcal{Y})$ such that

$$D_s\chi(\pi) = \mathbb{E}_{P_0}[\psi^\pi s],$$

defining the **efficient influence function (EIF)** as the Riesz representer of the pathwise derivative.



Left: $s(Z) = \partial_\varepsilon \log p_\varepsilon(Z)|_0$ is the slope of the log-density; it specifies the direction of infinitesimal change in P_ε.



Right: The parameter path $\varepsilon \mapsto \chi_\varepsilon(\pi)$ in $\mathcal{H}_\mathcal{Y}$; its tangent at 0 is $\mathbb{E}[\psi^\pi s]$, defining the EIF as the functional "gradient" of $\chi(\pi)$ with respect to the data law.

Appendix: The Efficient Influence Function (EIF) for the CPME – II

Influence functions. An *influence function* $h(Z) \in L^2(P_0; \mathcal{H}_\mathcal{Y})$ characterizes how a parameter $\chi(P)$ reacts to infinitesimal perturbations of the data law:

$$\frac{d}{d\varepsilon}\chi(P_\varepsilon)\Big|_0 = \mathbb{E}_{P_0}[h(Z)s(Z)], \qquad \mathbb{E}_{P_0}[h(Z)] = 0.$$

It quantifies how each observation Z contributes to the change in $\chi(P)$ when P is locally perturbed along the *score direction* $s(Z)$.

Why the tangent space. Let $\mathcal{T} \subseteq L^2(P_0)$ denote the **tangent space**—the closure of all valid score functions. It represents the set of directions in which the distribution P_0 can vary smoothly within the model. Projecting onto $\mathcal{T}$ ensures that we only consider directions that correspond to *real, estimable* perturbations of the data law.

Geometric interpretation. The **EIF** $\psi^\pi \in L^2(P_0; \mathcal{H}_\mathcal{Y})$ is the *unique element* whose inner product with any score reproduces the pathwise derivative:

$$D_s\chi(\pi) = \mathbb{E}_{P_0}[\psi^\pi s], \quad \forall s \in \mathcal{T}.$$

Equivalently, ψ^π is the *orthogonal projection* of any regular influence function h onto the tangent space $\mathcal{T}$—it isolates the part of h that corresponds to meaningful, model-consistent variation in P.

Appendix: The Efficient Influence Function (EIF) for the CPME – III

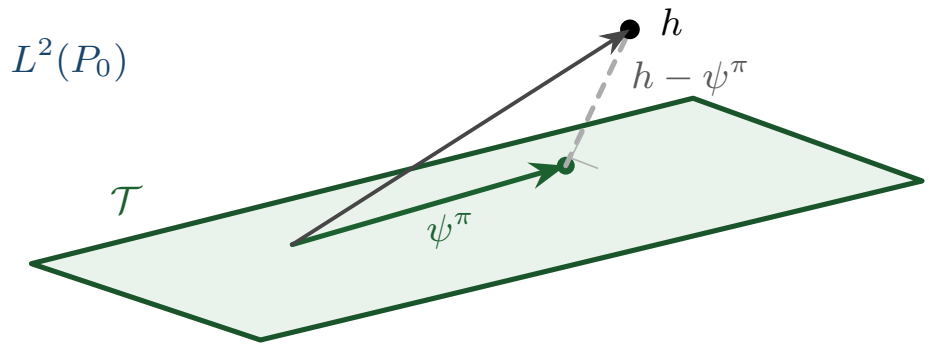


Figure: The EIF ψ^π is the orthogonal projection of any regular influence function h onto the tangent space $\mathcal{T} \subset L^2(P_0)$. The residual $h - \psi^\pi$ is orthogonal to $\mathcal{T}$, meaning that it corresponds to directions that cannot change $\chi(P)$.

Why it minimizes variance. Since $h - \psi^\pi$ is orthogonal to all valid scores:

$$\mathbb{E}_{P_0}[(h - \psi^\pi)s] = 0, \quad \forall s \in \mathcal{T},$$

the EIF ψ^π is the projection of h onto $\mathcal{T}$. By the Pythagorean theorem in $L^2(P_0)$,

$$\|h\|^2_{L^2(P_0)} = \|\psi^\pi\|^2_{L^2(P_0)} + \|h - \psi^\pi\|^2_{L^2(P_0)} \Rightarrow \|h\|^2_{L^2(P_0)} \geq \|\psi^\pi\|^2_{L^2(P_0)}.$$

Thus, the EIF yields the **smallest possible asymptotic variance**: it is the most efficient (shortest) direction in $L^2(P_0)$ that still preserves the correct local behavior of $\chi(\pi)$ under valid perturbations of P.